

Synaptic Pruning vs Sparse Firing: Why a Brain Stays Dynamic

20 Watts · Episode 4 — Synaptic Pruning

[Your Name], age 17 — June 2026 · github.com/svaka2000/20-watts

Abstract

The brain economizes in two different ways. Once, during development, it permanently **prunes** roughly half of its synapses (Huttenlocher, 1979) — a structural, one-time decision. And continuously, it keeps almost every neuron **silent**, firing fewer than 1% at a time — a dynamic, input-by-input decision (Episode 1). We measure the *static* kind on **Qwen2.5-7B-Instruct (4-bit)**: permanently remove the globally least-active MLP neurons (ranked by mean activation over a calibration set) and read held-out perplexity, then put it head-to-head with Episode 1's *dynamic* per-token sparsity at the same neuron counts. The contrast is stark. **Dynamic sparsity is free up to 60% skipping (+0.8% perplexity); the same fraction of static pruning costs +259%**. A fixed pruning that removes just 30% of neurons already costs +29%, while skipping a per-token-chosen 30% is free. Adaptivity is worth roughly **2× the sparsity**. The lesson — and the reason the brain bothers to keep its pruned-down circuits dynamic — is that *which* neurons a network needs is highly input-dependent: no single fixed subset can serve every token.

1. Introduction

Episode 1 showed a 7B model only needs ~40% of its feed-forward neurons on any given token, *if it is allowed to choose them per token*. A natural question is whether you even need that machinery: could you just delete the neurons that are rarely useful, once, and ship a permanently smaller model? That is **synaptic pruning** — the structural counterpart to sparse firing, and a real strategy in both neuroscience and ML (the Lottery Ticket Hypothesis, Frankle & Carbin 2019; SparseGPT, Frantar & Alistarh 2023; Wanda, Sun et al. 2023). Static pruning has a huge practical advantage: it is **free to realize** — a smaller weight matrix, no predictor, no special kernel. So the comparison matters: how much quality do you pay for that convenience?

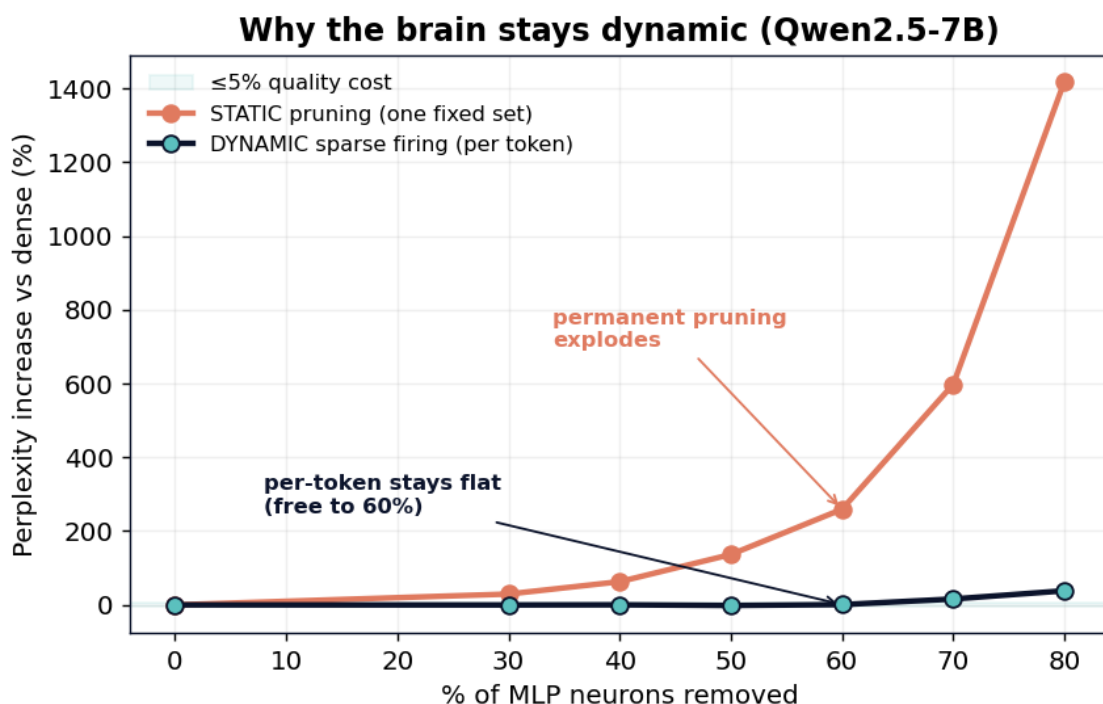
2. Method

Global importance. On a calibration corpus (WikiText-2, ~4k tokens) we record each MLP neuron's mean activation magnitude, `importance_l[j] = mean_t |h_l[j]|` where $h = \text{SiLU}(\text{gate}(x)) \odot \text{up}(x)$. This is an *activation-aware* importance (in the spirit of Wanda), not naive weight magnitude.

Static pruning. For a keep fraction p , we build one fixed per-layer mask keeping the top- p neurons by global importance, and apply it to **every** token. **Dynamic sparsity** keeps, per token, the top- p neurons by that token's own $|h|$ (Episode 1). Both use the same bit-exact harness (integrity diff = 0 at $p=1$). We report held-out perplexity on an authored passage not in the calibration set.

3. Results

Neurons skipped	Dynamic (per-token) Δ ppl	Static (global prune) Δ ppl
0%	— (ppl 24.09)	—
30%	−0.2%	+29.4%
40%	+0.2%	+63.0%
50%	−1.6%	+136.6%
60%	+0.8%	+259.4%
70%	+16.0%	+598.2%
80%	+38.3%	+1419%



At a 5%-perplexity budget, **dynamic affords 60% sparsity; static affords 0%**. Removing even the *globally least-active* 30% of neurons permanently is far more damaging than skipping a per-token-chosen 60%. The neurons that look idle on average are, for some specific tokens, essential.

4. Discussion

The value of adaptivity, quantified. The $\sim 2\times$ sparsity gap is the price of committing to one fixed circuit. It is also a clean explanation for the biology: a brain prunes to remove what it *never* needs, then keeps the survivors dynamic because what it needs *right now* keeps changing. Pruning and firing are not redundant; they are sequential.

An honest caveat. Our static baseline uses activation-aware global importance — a reasonable metric, but not the state of the art. SparseGPT and Wanda use second-order or weight $\times$ activation criteria and

would shrink the gap, especially at low sparsity. They cannot, however, escape the fundamental limit demonstrated here: a single fixed subset cannot be optimal for inputs that need different subsets. The *direction* of the result is robust even if the exact magnitudes would soften under a stronger pruner.

What stacks. Static pruning composes with quantization (a smaller *and* coarser model) and is the most deployment-friendly lever (no runtime machinery). Dynamic sparsity buys more, but only with the predictor of Episode 1. The efficient model of the future is plausibly **pruned once, quantized, and then run sparsely** — all three.

5. Conclusion

Given the same 7B model, dynamic per-token sparsity tolerates roughly twice the neuron removal of static pruning before quality breaks. The brain's two economies are not interchangeable: prune what you never need, but stay dynamic about what you need now. Episode 4 closes the loop opened in Episode 1 — and measures, in perplexity, why the brain is dynamic at all.

References

1. P. R. Huttenlocher. *Synaptic density in human frontal cortex — developmental changes and effects of aging*. Brain Research, 1979.
2. J. Frankle & M. Carbin. *The Lottery Ticket Hypothesis*. ICLR, 2019.
3. E. Frantar & D. Alistarh. *SparseGPT: Massive Language Models Can Be Accurately Pruned in One-Shot*. ICML, 2023.
4. M. Sun, Z. Liu, A. Bhojanapalli, et al. *A Simple and Effective Pruning Approach for LLMs (Wanda)*. 2023.
5. Z. Li et al. *The Lazy Neuron Phenomenon*. ICLR, 2023.
6. Z. Liu et al. *Deja Vu: Contextual Sparsity for Efficient LLMs*. ICML, 2023.